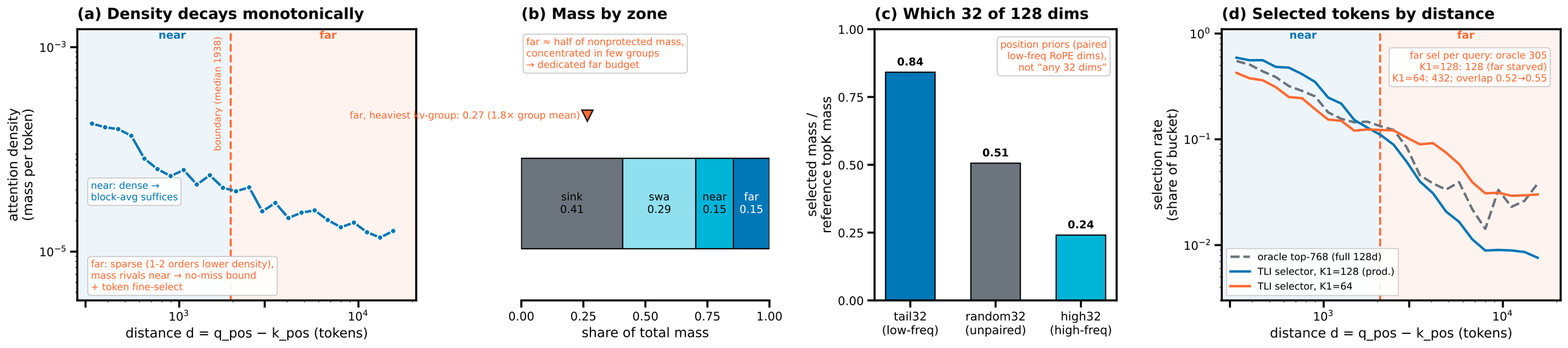
(a) Density decays monotonically
near
far
boundary (median 1938)
near: dense →
block-avg suffices
far: sparse (1-2 orders lower density),
mass rivals near → no-miss bound
+ token fine-select
attention density
(mass per token)
distance d = q_pos − k_pos (tokens)
(b) Mass by zone
far ≈ half of nonprotected mass,
concentrated in few groups
→ dedicated far budget
far, heaviest kv-group: 0.27 (1.8× group mean)
sink
0.41
swa
0.29
near
0.15
far
0.15
share of total mass
(c) Which 32 of 128 dims
position priors (paired
low-freq RoPE dims),
not “any 32 dims”
0.84
0.51
0.24
selected mass /
reference topK mass
tail32
(low-freq)
random32
(unpaired)
high32
(high-freq)
(d) Selected tokens by distance
near
far
far sel per query: oracle 305
K1=128: 128 (far starved)
K1=64: 432; overlap 0.52→0.55
oracle top-768 (full 128d)
TLI selector, K1=128 (prod.)
TLI selector, K1=64
selection rate
(share of bucket)
distance d = q_pos − k_pos (tokens)